

A Novel Task-Driven Diffusion-Based Policy with Affordance Learning for Generalizable Manipulation of Articulated Objects

Hao Zhang, Zhen Kan, Weiwei Shang, and Yongduan Song

Abstract—Despite recent advances in dexterous manipulations, the manipulation of articulated objects and generalization across different categories remain significant challenges. To address these issues, we introduce DART, a novel framework that enhances a diffusion-based policy with affordance learning and linear temporal logic (LTL) representations to improve the learning efficiency and generalizability of articulated dexterous manipulation. Specifically, DART leverages LTL to understand task semantics and affordance learning to identify optimal interaction points. The diffusion-based policy then generalizes these interactions across various categories. Additionally, we exploit an optimization method based on interaction data to refine actions, overcoming the limitations of traditional diffusion policies that typically rely on offline reinforcement learning or learning from demonstrations. Experimental results demonstrate that DART outperforms most existing methods in manipulation ability, generalization performance, transfer reasoning, and robustness. For more information, visit our project website at: <https://sites.google.com/view/dart0257/>.

I. INTRODUCTION

The manipulation of articulated objects has been an interesting and important topic in robotic learning. Although prior research has demonstrated promising results in the manipulation of rigid bodies, significant challenges persist when it comes to handling articulated objects [1]. Generalizing to various types of articulated objects [2] is particularly difficult for dexterous manipulations. For example, if a dexterous hand can open the lid of a toilet, it should also be capable of opening the lid of a garbage can, despite their cosmetic differences. While many recent efforts have focused on improving the robotic generalization performance [3] or reducing the exploration burden [4], enhancing the learning efficiency or improve the generalization ability for high degrees of freedom (DOF) skills, such as dexterous manipulation, remains a challenging problem, not to mention achieving both simultaneously.

There are various methods enabling skill generalization via reinforcement learning (RL). Taking advantage of Meta-RL to improve the generalization potential of algorithms is an effective approach [5]. However, solving context Markov decision processes (MDPs) in Meta-RL often involves

cumbersome computation. Recently, increasing efforts are being dedicated to leveraging vision features to enhance RL in tackling various challenging tasks [6]–[8]. One of the interesting findings is that algorithms exploiting point cloud information can have good generalization without any Meta-RL-like training [9]. However, most of the previous work struggles to understand the geometry of articulated objects, thus making it challenging to identify appropriate interaction points when developing manipulation skills.

One of the effective ways to address the above shortcomings is affordance learning [10]. In [11], a learning-from-interaction framework was proposed to manipulate articulated objects, which uses an actionability scoring module to estimate the score of each pixel and provide better interaction positions. The work of [12] further extends [11] by exploiting the multi-stage stable learning and self-supervised data collection to achieve deformable object manipulation. In [13], RL was guided by affordance learning to achieve diverse manipulation, which yields an end-to-end learning framework to avoid offline data collection. Although affordance learning has shown great potential in the manipulation of articulated objects, generating a high-quality affordance map often requires a significant amount of interactions for training, which significantly impacts learning efficiency in long-horizon tasks.

Due to the rich expressivity, linear temporal logic (LTL) [14] is capable of describing long-horizon tasks composed of logically organized sub-tasks. For instance, LTL specification was converted to a finite-state predicate automaton in [15] to facilitate the use of RL in performing complex manipulation tasks. However, previous methods mainly focus on solving tasks with different automaton, without considering the task semantics of LTL instructions. Based on graph neural networks, an approach exploiting the compositional syntax and semantics of LTL was proposed in [16] to learn task-conditioned policies, which is generalizable to new instructions. In [17], Transformer [18] was employed to encode the LTL specifications for improved representation capability and interpretability. Direct language approaches that utilize large language models [19] or represent tasks using text [20] have brought success to manipulation at the task level. However, these approaches are limited either by their expressiveness and rigor, or by the computational efficiency of the algorithms relative to LTL. Despite recent progress, neither vision-based RL nor LTL-guided motion planning inherently improves the generalization of the learned manipulation skills.

As an effective generative model, the diffusion model (DM)

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[21]–[23] utilizes a parametric reverse process to generate samples from a Gaussian distribution. Inspired by this, DM has been investigated for long-horizon manipulation [24]–[26]. However, in most of these studies, DM has functioned as a policy within either an offline reinforcement learning or imitation learning framework. Extending this approach to dexterous manipulations is challenging due to the difficulty in obtaining high-quality operational data. Our primary motivation is to determine whether task semantics can be leveraged to drive affordance learning and guide the agent in achieving a policy with the diffusion model, thereby improving learning efficiency for articulated objects and enhancing generalization in operational performance. To address this, we propose DART, a novel policy generalization framework that enhances a diffusion-based policy using affordance learning and linear temporal logic representations to improve both generalizability and learning efficiency. DART guides the agent in improving sampling efficiency by encoding task semantics into LTL representations. Additionally, it utilizes affordance learning to generate an affordance map based on point cloud information captured by the camera, predicting optimal contact points for dexterous manipulation. Furthermore, DART incorporates a task-driven action optimization method for the policy, enabling training and optimization in online settings.

Contributions: The main contributions of this work are summarized as follows:

- We introduce a novel policy generalization algorithm that enhances a diffusion-based policy through the use of LTL representations and affordance learning, thereby improving learning efficiency and ensuring robust generalization performance.
- We construct a task-driven MDP based on LTL representations, which not only incorporates task semantics to facilitate robotic learning but also designs a contact planner based on affordance learning to enhance the efficiency of articulated manipulations.
- We exploit a diffusion-based policy for interacting with articulated objects, whose multimodal policy representation lays the groundwork for the generalization.
- We further extend the task environment of [9] and evaluate the methods when generalizing to different category-level objects. A comprehensive empirical study validates that DART significantly outperforms baseline methods in terms of manipulation ability, generalization performance, transfer reasoning, and robustness.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. sc-LTL and LTL Progression

Co-safe LTL (sc-LTL) is a subclass of LTL that can be satisfied by finite-horizon state trajectories [27]. Since sc-LTL is suitable for describing robotic tasks, this work focuses on sc-LTL. An sc-LTL formula is built on a set of atomic propositions Π that can be true or false, standard Boolean operators such as $\wedge$ (conjunction), $\vee$ (disjunction), and $\neg$ (negation), temporal operators such as $\bigcirc$ (next), $\diamond$ (eventually), and $\cup$ (until). The semantics of an sc-LTL formula are interpreted over a word $\sigma = \sigma_0\sigma_1\dots\sigma_n$, which

is a finite sequence with $\sigma_i \in 2^\Pi$, $i = 0, \dots, n$, where 2^Π represents the power set of Π . Denote by $\langle \sigma, i \rangle \models \varphi$ if the sc-LTL formula φ holds from position i of σ .

LTL formulas can also be progressed along a sequence of truth assignment [28]. Specifically, give an LTL formula φ and a word $\sigma = \sigma_0\sigma_1\dots$, the LTL progression $\text{prog}(\sigma_i, \varphi)$ at step i is $\text{prog}(\sigma_i, p) = \text{True}$ if $p \in \sigma_i$, where $p \in \Pi$ and $\text{prog}(\sigma_i, p) = \text{False}$ otherwise. The operator prog takes an LTL formula φ and the current label σ_i as input at each step, and outputs a formula indicating the remaining instructions to be addressed.

B. Labeled MDP and Visual Reinforcement Learning

When performing the sc-LTL task φ in visual RL, the interaction between the robot and the environment can be modeled by a labeled MDP $\mathcal{M}_e = (S, T, A, p_e, \Pi, L, R, \gamma, \mu, \Omega, O)$, where S is the state space, $T \subseteq S$ is a set of terminal states, A is the action space, $p_e(s'|s, a)$ is the transition probability from $s \in S$ to $s' \in S$ under action $a \in A$, Π is a set of atomic propositions indicating the properties associated with the states, $L : S \rightarrow 2^\Pi$ is the labeling function, $R : S \rightarrow \mathbb{R}$ is the reward function, $\gamma \in (0, 1]$ is the discount factor, μ is the initial state distribution, Ω is the observation space, and $O : S \rightarrow \Omega$ is the observation function that maps an environment state s to an observation in Ω . The labeling function L can be seen as a set of event detectors that trigger when $p \in \Pi$ presents in the environment, allowing the robot to determine whether or not an LTL specification is satisfied.

For any task φ , the robot interacts with the environment following a deterministic policy π_e over $\mathcal{M}_e$, i.e., given a state s_t at time t , $\pi_e : \Omega \rightarrow A$ maps an observation $O(s_t)$ to an action $a \in A$. The robot then receives a reward $r_t = R(s_t)$. Given a task φ , the goal of the agent is to learn an optimal policy $\pi^*(a|o)$ that maximizes the expected discounted return $E \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid O_t = o \right]$ given an observation $o \in O$.

C. SDE in Diffusion Model

Diffusion model [21]–[23], as a generative model, transforms any data distribution into a simple Gaussian distribution by adding noise (forward process) and then denoise it using neural networks (reverse process) with parameterized estimators. The forward noising process in continuous time, denoted as dx_t , add noise to the data $x \in \mathbb{R}^p$ following the Stochastic Differential Equation (SDE)

$$dx_t = \mathbf{f}(x_t, t) dt + g(t) d\mathbf{w}_t \quad (1)$$

where $\mathbf{f}(\cdot, t) : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is a Lipschitz function, $\mathbf{w}$ is a standard Brownian motion and $g(t) \in \mathbb{R}$ is a noise schedule which injects a fixed series of noise levels monotonically increasing via forward diffusion t . The reverse denoising process, denoted as $d\bar{x}_t$, generates new samples by utilizing the results of [23] for sampling

$$d\bar{x}_t = [\mathbf{f}(\bar{x}_t, t) - g^2(t) \nabla \log p_t(\bar{x}_t)] dt + g(t) d\bar{\mathbf{w}}_t \quad (2)$$

where $\nabla \log p_t(\cdot)$ represents the gradient of the logarithm of the marginal density function of the data x_t and $\bar{\mathbf{w}}$ denotes a reverse Brownian motion. It is proposed in [22] to utilize the noise estimator $\varepsilon_\xi(\bar{x}_t, t)$ as the score of distribution of $\bar{x}_t$ to approximate $\nabla \log p_t(\bar{x}_t)$ so that (2) can be rewritten as

$$d\bar{x}_t = [\mathbf{f}(\bar{x}_t, t) - g^2(t) \varepsilon_\xi(\bar{x}_t, t)] dt + g(t) d\bar{\mathbf{w}}_t.$$

In this work, we will replace the above data x with action a to design the decision-making process in RL using the diffusion-based policy for better generalization to different tasks.

D. Challenges and Problem Formulation

To elaborate the problem and challenges, the following running example will be used throughout the work.

Example 1. Consider a dexterous manipulation [9] as shown in Fig. 1(a), in which the robot needs to approach the toilet and open the lid O_{lid} . The set of propositions Π is {toilet_approached, lid_grasped, lid_opened}. Using above propositions in Π , an example sc-LTL formula is $\varphi_{\text{toilet}} = \Diamond(\text{toilet_approached} \wedge \Diamond(\text{lid_grasped} \wedge \Diamond \text{lid_opened}))$, which requires the robot to sequentially approach, grasp, and open the lid of the toilet.

There are three main challenges in Example 1. The first challenge is how to understand the task semantics and thus improve the learning efficiency. The second challenge is how to predict the optimal contact points for dexterous manipulation of articulated objects based on point cloud information. The third challenge is how to generalize the learned manipulation to different objects of the same kind, or even different category-level objects. To this end, we aim to develop a trainable and generalizable policy π_φ that can overcome these challenges. Hence, the problem can be presented as follows.

Problem 1. Given a labeled MDP $\mathcal{M}_e = (S, T, A, p_e, \Pi, L, R, \gamma, \mu, \Omega, O)$ corresponding to task φ with the reward function $R_\varphi(s_0 s_1 \dots s_t)$ to be designed, the goal of this work is to find an optimal policy π_φ^* over the LTL instruction φ , so that the successful rate evaluated on seen and unseen sets of articulated objects under the policy π_φ^* can be maximized.

III. MANIPULATION GENERALIZATION

In this section, we present a novel framework, namely task-driven diffusion-based policy with affordance learning for generalizable manipulation of articulated objects (DART), that enables the agent to learn efficiently and generalize diversely. We first describe the overview of DART in Sec. III-A and then explain the technical details in Sec. III-B to Sec. III-E.

A. Overview of DART

As shown in Fig. 1, DART uses LTL representations and affordance learning to guide the diffusion-based policy to improve the learning efficiency and the articulated dexterous generalization. Since traditional diffusion policies highly rely on demonstrations, a task-oriented optimization approach is incorporated for action improvement. The developed DART

has several advantages. First, by extracting the task semantics of LTL formula via Transformer encoder, the agent can more efficiently explore and learn the dexterous manipulation. Second, instead of simply feeding visual features to the RL module as in literature, we predict a score map through affordance learning and plan an optimal manipulation point to guide the interaction with the articulated objects. In addition, we exploit a diffusion-based policy that can be learned from scratch, which not only inherits and promotes the generalization of the diffusion model itself, but also makes use of the action improvement to enhance the performance.

By exploiting the LTL progression introduced in Sec. II-A, we develop an augmented MDP with an LTL task φ , namely task-driven labeled MDP (TL-MDP), as follows.

Definition 1. Given a labeled MDP $\mathcal{M}_e = (S, T, A, p_e, \Pi, L, R, \gamma, \mu, \Omega, O)$ corresponding to an LTL task φ , the TL-MDP is constructed by augmenting $\mathcal{M}_e$ to $\mathcal{M}_\varphi \triangleq (\tilde{S}, \tilde{T}, A, \tilde{p}, \Pi, L, \tilde{R}_\varphi, \gamma, \mu, \Omega, O)$, where $\tilde{S} = S \times \text{cl}(\varphi)$ with $\text{cl}(\varphi)$ denoting the progression closure of φ , i.e., the smallest set containing φ that is closed under progression, $\tilde{T} = \{(s, \varphi) | s \in T, \text{or } \varphi \in \{\text{True}, \text{False}\}\}$, $\tilde{p}((s', \varphi') | (s, \varphi), a) = p_e(s' | s, a)$ if $\varphi' = \text{prog}(L(s), \varphi)$ and $\tilde{p}((s, \varphi') | (s, \varphi), a) = 0$ otherwise, and $\tilde{R}_\varphi$ is the reward function associated with the task φ to overcome the non-Markovian reward issue which is designed as

$$\tilde{R}_\varphi(s, \varphi) = \begin{cases} r_{\text{env}} + r_\varphi, & \text{if } \text{prog}(L(s), \varphi) = \text{True}, \\ r_{\text{env}} - r_\varphi, & \text{if } \text{prog}(L(s), \varphi) = \text{False}, \\ r_{\text{env}}, & \text{otherwise.} \end{cases} \quad (3)$$

where r_{env} is the environmental reward, and r_φ corresponds to the extra reward when σ satisfies the LTL task φ .

The pseudo-code of DART is shown in Alg. 1, which includes an exploration phase (lines 4-11) and a training phase (lines 12-16). In the exploration phase, affordance learning is first used to predict the max-affordance point of manipulation based on the point clouds of the object at the initial moment (lines 6-7). Meanwhile, LTL representations are encoded using Transformer (line 8). During the interaction, the diffusion-based policy outputs an action a to interact with the environment guided by vision features and LTL representations (line 9). Then the transition corresponding to the action a is added to the data set $\hat{B}$ (line 10). In the training phase, the transitions are not only sampled to update the critic and the diffusion-based policy, but also fed to the noise estimator to fit the approximate $\nabla \log p_t(\cdot)$ (lines 13-14). In addition, the state-action pairs (s, a) in $\hat{B}$ are exploited to optimize the action towards the reward improvement (line 14). When the manipulation success rate exceeds the threshold, the contact planner is updated with the contact positions to predict better interaction points (line 15). Note that the task representation module, which incorporates the semantics of the LTL instruction φ and encodes the LTL representation, can be updated indirectly by the gradient backpropagation of RL module. This process continues until the training budget is exhausted. Then, DART terminates and returns the learned policy and network parameters.

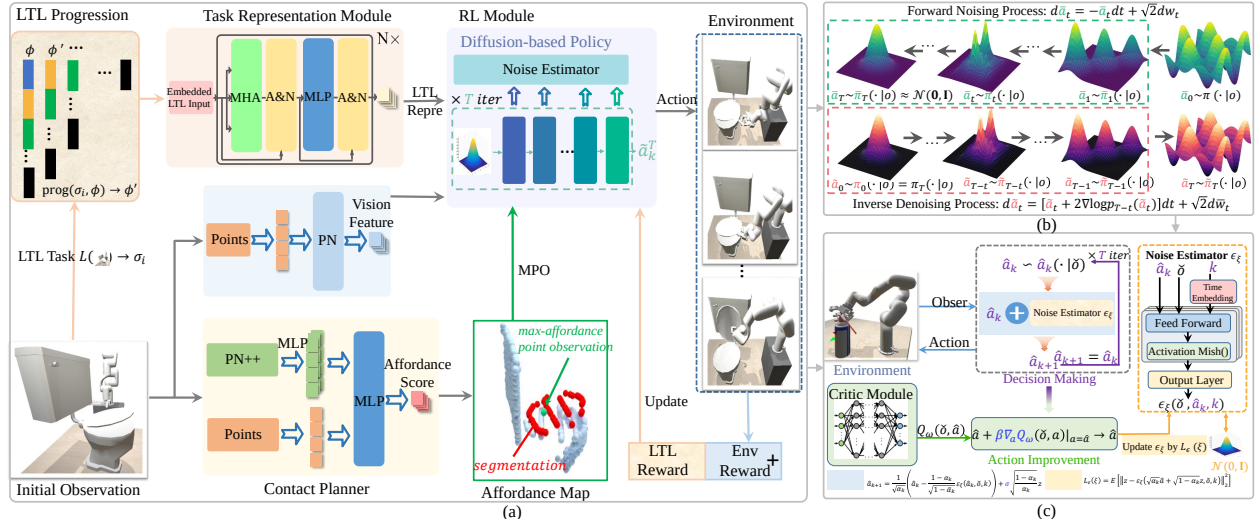


Fig. 1. (a) The framework of DART which consists of three key modules. Task representation module: the Transformer encoder architecture is employed to encode LTL instruction φ to facilitate the understanding of task semantics. Contact Planner: affordance learning is applied to generate an affordance map based on observation and predict the MPO. RL Module: A diffusion-based policy is used to interact with the objects and improve the generalization performance. (b) The outlines of a series of forward noise addition policy distributions and a series of inverse denoising policy distributions. (c) The outline of a diffusion-based policy and action improvement.

Algorithm 1 DART

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1: procedure INPUT: (An LTL instruction  $\varphi$  and the MDP  $\mathcal{M}_e$  corresponding to some  $\varphi$ )
   Output: Improved action  $\hat{a}_K$ , optimized Contact Planner  $CP(\cdot)$ , noise estimator  $\varepsilon_\varepsilon(\cdot)$  and Transformer Encoder in TL-MDP  $\mathcal{M}_\varphi$ 
   Initialization: All neural network weights
2: Load the pretrained weights to the PointNet  $PN(\cdot)$ 
3: while episode not terminated do
4:   for step  $t = 0, 1, \dots, T$  do ▷ Exploration Phase
5:      $\varphi' \leftarrow \text{prog}(L(s), \varphi)$ 
6:     Extract the vision representation  $o_{pn}$  from 3D point cloud by the PointNet  $PN(\cdot)$ 
7:     Get the max-affordance point observation  $o_{MP}$  from object point clouds  $o^{obj}$  by the contact planner  $CP(\cdot)$ 
8:     Augment the observation  $o_0$  with  $\varphi_\theta$  encoded by Transformer
9:     Determine  $\hat{R}_\varphi$  by (3) and gather data from  $\varphi$  following the diffusion-based policy  $\pi_\varphi$ 
10:    Add the corresponding transition to the data set  $\hat{B}$ 
11:   end for
12:   for training step  $t = 0, 1, \dots, K$  do ▷ Training Phase
13:     Update the noise estimator  $\varepsilon_\varepsilon(\cdot)$  in diffusion process by (7)
14:     Improve the action performance by (8) and update the corresponding critic module by (9)
15:     Optimize Contact Planner  $CP(\cdot)$  by (4)
16:   end for
17: end while
18: Get Optimized neural network weights
19: end procedure

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In the following, Sec. III-B describes the extraction of point cloud features and the complementation of the agent's point cloud. Sec. III-C presents how Transformer extracts the task semantics of the LTL formula and guides the agent to explore the environment. Sec. III-D depicts a series of designs in DART around affordance learning to improve the success rate of articulated manipulation. Sec. III-E explains how the online diffusion-based policy fits the reverse process and performs the action improvement.

B. Feature Extraction

Previous works typically focus on using images as visual inputs to guide the manipulation of articulated objects [7].

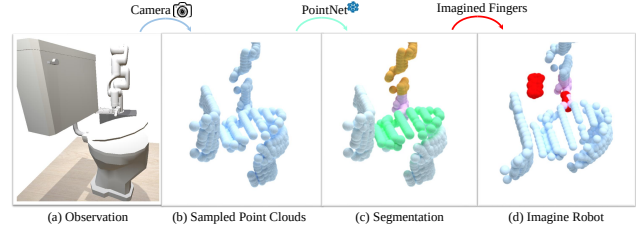


Fig. 2. A series of processes on (a) observation containing (b) point cloud sampling, (c) segmentation, and (d) occluded fingers imagination. The green color represents the functional part of the toilet in (c), and the red color shows the imagined fingers in (d). The snowflake indicates the frozen weights.

However, images often lack necessary spatial information and are vulnerable to perspective changes, resulting in degraded performance. Instead, the visual features extracted from 3D point clouds contain spatial information that can effectively improve the generalization of the agent's performance and provide a robust basis for the object manipulation under different camera viewpoints. Hence, this work is motivated using 3D point clouds to extract the corresponding visual features to guide the manipulation.

Specifically, we use PointNet [29] $PN(\cdot)$ to extract vision features o_{pn} from 3D point clouds. Since it is difficult to make PointNet output stable using the gradient backpropagation of RL controller, we construct a dataset to pre-train the PointNet's weights and freeze its weights when using with RL. In this work, the PointNet is pre-trained by the segmentation method and the extracted visual features allow the agent to make basic inferences about the functional parts of articulated objects. For this purpose, we categorize the operational scenarios of articulated objects into the following four groups, i.e., the functional part of the object, the rest of the object, the dexterous hand, and the robot arm, which are labeled and used for the training of PointNet.

However, there are two main issues associated with point clouds. One issue is that the occlusion of finger point clouds during interaction may affect the efficiency of learning and the stability of performance. Another issue is that, when sampling as many points as possible to avoid loss of important point clouds, the high memory usage can seriously increase the training time and resource consumption. To avoid the issues above, we utilize imagined point clouds to compensate for occluded finger point clouds even if the information is from a single camera view. Specifically, we first calculate the pose of each finger via forward kinematics based on the robotic kinematics model and joint positions. Then, we utilize the potential pose to sample points from the mesh of each finger link to synthesize the imagined point cloud of the corresponding finger as shown in Fig. 2.

C. Task Representation Module based on LTL

In this section, we focus on the use of LTL representations to understand the task semantics to improve agent's performance. As shown in Fig. 1(a), LTL progression is first utilized to progress the LTL instruction. The operator prog mentioned in Sec. II-A takes an LTL formula φ and the current label σ_i as input at each step, and outputs a formula to track which parts of the original instructions remain to be addressed. For instance, consider the task φ_{toilet} in Example 1, which will progress to the subsequent sub-task $\varphi'_{\text{toilet}} = \Diamond(\text{lid_grasped} \wedge \Diamond \text{lid_opened})$ as soon as the robot reaches toilet_approached . We then employ Transformer to encode LTL representation as part of the input to the RL module, which can indirectly update the task representation module using the gradient backpropagation of the reward obtained from the interaction.

Instead of converting LTL specifications to automata or Reward Machine, our previous work [17] also shows the representation encoded by Transformer [18], which provides flexibility in encoding LTL instructions and facilitates the agent's training. Specifically, given an input $X_\varphi = (x_0, x_1, \dots)$ generated by the LTL task φ where $x_t, t = 0, 1, \dots$, represents the operator or proposition, X_φ will be preprocessed by the word embedding E as $X_E = [x_0E; x_1E; \dots; x_NE] \in \mathbb{R}^{B \times M \times D}$ where B is the batch size, $M = N + 1$ is the length of input X_φ and D is the model dimension of the Transformer. And then X_E is added with the frequency-based positional embedding E_{pos} to make use of the order of the sequence. The architecture of Transformer Encoder for DART is shown in Task Representation Module of Fig. 1(a). Incorporating the LTL representation in TL-MDP is not only appropriate for the forward propagation of the neural network, but also can be continuously updated as Transformer evolves.

D. Contact Planner based on Affordance Learning

To achieve generalizable dexterous manipulation, it is crucial to understand how to effectively manipulate the functional part. However, to achieve human-level intelligence, previous methods involve either a large number of random contacts in the offline phase to obtain effective sampling [11], or human remote control to calibrate reasonable interaction positions [30], which greatly increases the learning cost. A

feasible approach was proposed in [31], where the affordance module is trained using human demonstrations. However, collecting such demonstrations for different objects is labor-intensive. Moreover, since the manipulated object in this work is articulated, visualizing the contact posture of all five fingers is challenging, unlike in the case of rigid bodies where finger contact can be easily observed. This difficulty further hinders the efficient learning of the affordance module.

To address these challenges, we design a contact planner $\text{CP}(\cdot)$ that eliminates the need for offline random acquisition or human demonstrations. Additionally, we develop an effective affordance module that can be optimized through end-to-end online interactions. Instead of fitting contact postures directly, our approach transforms the predicted five-finger contact gesture into a predicted max-affordance point observation (MPO), which guides the palm's interaction. Specifically, the contact planner is mainly composed of PointNet++ [32] and actionable affordance module, where PointNet++ mainly extracts visual features from the segmented point clouds of the object o^{obj} , while the actionable affordance module further maps the visual features into actionability scores of each point to indicate the different potential for successful interaction, thus converting the point cloud of an object into a corresponding affordance map. Based on the affordance map, we select the top K points with the highest actionable scores and average their positions to obtain the MPO which represents the interaction point with the highest success rate. The agent then uses the MPO as additional information o_{MP} , which is inputted into the RL controller, to guide the agent's end-effector s_{eff} (e.g., the palm position of the dexterous hand) to interact with the articulated object, thus achieving effective generalization. In addition, to encourage the end-effector s_{eff} to manipulate the functional part of the articulated object as close as possible to the MPO o_{MP} , we design the max-affordance point reward (MPR), $r_{\text{MPR}} = 1 / \|o_{\text{MP}} - s_{\text{eff}}\|$, as a part of r_{env} to indirectly influence the agent's decision-making.

Through the interaction between the agent and the environment, the interaction position at which its end-effector completes the task will further provide the optimized direction for the contact planner's output. Specifically, during the interaction, we record the contact points of the end-effector s_{eff} as $\text{DGT}^i(s_{\text{eff}})$ and the corresponding success rates sr^i when the articulated object i is manipulated in a fixed number of steps. Using the success rate sr^i as a scale for updating, the following process was designed to efficiently update and learn the following optimal contact planner

$$\text{CP}^* = \underset{\text{CP}}{\text{argmin}} \sum_i sr^i \left\| \sum_{o_{\text{MP}}} \text{CP}(o_{\text{MP}} | o_i^{obj}) - \text{DGT}^i(s_{\text{eff}}) \right\|_2. \quad (4)$$

E. Diffusion Model as Policy Representation for RL

While the visual features extracted in Sec. III-B can provide a basic reasoning ability for the agent's generalization performance, the quality of the interaction actions is one of the key factors influencing its manipulation results. Exploiting

diffusion model as a policy representation can improve the generalization performance of the agent, which has been proven successful in areas such as manipulation. Exploiting diffusion model as a policy representation can improve the generalization performance of the agent, which has been proven successful in areas such as manipulation [24]–[26]. However, the generalizability of most diffusion model-based methods in robotics relies on either solving the problem with offline reinforcement learning (RL) or learning from demonstrations. Such a setting is hard to be applied for dexterous manipulation as related demonstrations are difficult and expensive to collect. Therefore, in this section, we propose a diffusion-based policy without prior in the context of RL, whose unique action improvement not only addresses the above challenges but also ensures generalization.

To simplify the computation mentioned in Sec. II-C, we consider $\mathbf{f}(a_t, t) = -\frac{1}{2}g^2(t)a_t$ with $g(t) = \sqrt{2}$ in (1) and (2), whose forward noise addition and inverse denoising processes can be rewritten as $d\bar{a}_t = -\bar{a}_t dt + \sqrt{2}d\mathbf{w}_t$ and

$$d\tilde{a}_t = [\tilde{a}_t + 2\nabla \log p_{T-t}(\tilde{a}_t)]dt + \sqrt{2}d\tilde{\mathbf{w}}_t. \quad (5)$$

Thus, as shown in Fig. 1(b), we can derive $\bar{a}_t$ from a series of forward noise addition policy distributions $\{\pi_t(\cdot|o)\}_{t=0:T}$, where $\pi_0(\cdot|o) = \pi(\cdot|o)$ is the input policy distribution and $\pi_T(\cdot|o)$ is standard Gaussian distribution. Similarly, $\tilde{a}_t$ can be obtained from a series of inverse denoising policy distributions $\{\tilde{\pi}_t(\cdot|o)\}_{t=0:T}$ in (5), where $\tilde{\pi}_t(\cdot|o) = \pi_{T-t}(\cdot|o)$ if $\tilde{a}_0 \sim \pi_T(\cdot|o)$ and $\nabla \log p_t(\bar{a}_t) = \nabla \log \tilde{\pi}_t(\tilde{a}_t|o)$. For any visual features o_{pn} corresponding to observation o , LTL representation φ_θ , the max-affordance point observation o_{MP} and proprioceptive data s_{prop} , we can design the noise estimator $\varepsilon_\xi(\bar{a}_t, o_{pn}, \varphi_\theta, o_{MP}, s_{prop}, t)$ to approximate the gradient of the logarithm of the marginal density function $\nabla \log \pi_t(\bar{a}_t|o_{pn}, \varphi_\theta, o_{MP})$ in the dexterous manipulation by

$$L_\varepsilon(\xi) = \int_0^T \omega(t) \frac{E}{\bar{a}_0 \sim \pi(\cdot|o), \bar{a}_t | \bar{a}_0} [\|\varepsilon_\xi(\bar{a}_t, o_{pn}, \varphi_\theta, o_{MP}, s_{prop}, t) - \nabla \log \psi_t(\bar{a}_t | \bar{a}_0)\|_2^2] dt$$

where $\omega(t) : [0, T] \rightarrow \mathbb{R}^+$ means a positive weight function, $\psi_t(\bar{a}_t | \bar{a}_0)$ represents the transition kernel of the forward process (1), and ξ are parameters in noise estimator ε_ξ . For brevity, we denote the tuple of visual features o_{pn} , LTL representation φ_θ , the max-affordance point observation o_{MP} and proprioceptive data s_{prop} by $\check{o}$ as $\check{o} = (o_{pn}, \varphi_\theta, o_{MP}, s_{prop})$. When the agent interacts with the environment, we can obtain the exact solution from the reverse denoising process (5) by performing K iterations to gradually denoise a random noise $\hat{a}_0$ into the noise-free action $\hat{a}_K$ as

$$\hat{a}_{k+1} = \frac{1}{\sqrt{\alpha_k}} \left(\hat{a}_k - \frac{1 - \alpha_k}{\sqrt{1 - \alpha_k}} \varepsilon_\xi(\hat{a}_k, \check{o}, k) \right) + \sigma \sqrt{\frac{1 - \alpha_k}{\alpha_k}} z \quad (6)$$

where $z \sim \mathcal{N}(0, \mathbf{I})$, $\sigma \in \mathbb{R}$, $\alpha_k \in \mathbb{R}$, and $\bar{\alpha}_k = \prod_{i=1}^k \alpha_i$ are coefficients dependent on the iteration k . Then we can add the corresponding transition to the data set $\hat{B}$. Note that $T = hK$ where $h \in \mathbb{Z}^+$ is the step size to discrete the interval $[0, T]$, and $\sigma = 0$ when deploying and $\sigma = 1$ otherwise.

Compared with the priority-based diffusion policy which only needs to update the estimator $\varepsilon_\xi(\cdot)$, when updating the diffusion-based policy in DART, we not only need to train the estimator $\varepsilon_\xi(\cdot)$ to approximate $\nabla \log p_t(\cdot)$, but also want to sample the good action a from the Gaussian distribution, so that it can fulfill manipulation tasks as soon as possible. Specifically, the update of the policy in DART can be divided into the following two steps.

1) Update the estimator $\varepsilon_\xi(\cdot)$: Inspired by [22], the loss $L_\varepsilon(\xi)$ can be simplified to

$$L_\varepsilon(\xi) = \frac{E}{(\check{o}, a) \sim \hat{B}} \left[\|z - \varepsilon_\xi(\sqrt{\alpha_k} \hat{a} + \sqrt{1 - \alpha_k} z, \check{o}, k)\|_2^2 \right]. \quad (7)$$

2) Action improvement of diffusion-based policy: In (6), since the inverse denoising process to obtain action $\hat{a}$ does not contain any policy parameters, traditional RL method is difficult to effectively update the agent's decision. Therefore, we incorporate the action improvement for the following two reasons. On one hand, from (6), it can be seen that in the K iterations, the action in the previous step plays an important role in the quality of the action in the next step. On the other hand, from the RL point of view, the measure of how good an agent's output action is the reward it receives when interacting with the environment. To summarize, we propose action improvement:

$$\hat{a} \leftarrow \hat{a} + \beta \nabla_a J(a) = \hat{a} + \beta \frac{E}{(\check{o}, a) \sim \hat{B}} [\nabla_a Q_\omega(\check{o}, \hat{a})], \quad (8)$$

where β is the learning rate and $Q_\omega(o, a)$ is the value function parameterized with weights ω whose update follows

$$J_Q(\omega) = \frac{E}{(\check{o}, a) \sim \hat{B}} \left[\left(Q_\omega(\check{o}, \hat{a}) - \left(\tilde{R}_\varphi + (\check{o}', \hat{a}') \right) \right)^2 \right]. \quad (9)$$

By designing (8), the agent not only overcomes the limitation of depending on the prior in the previous diffusion policy, but also takes advantage of multi-modal policy representation in diffusion process to provide a good basis for generalization. In addition, although DART does not involve the updating of policy parameters, the action improvement can ensure convergence, as demonstrated in [25].

F. Advantages Analysis

The advantages of DART can be summarized as follows.

First, compared with traditional visual RL algorithms, which are usually based on RGB-D images to guide the agent to perform manipulations, the extracted features affect the agent's spatial reasoning ability. Differently, we extract the visual features of the current scene based on point clouds via PointNet, and provide a generalization basis for bringing the agent to manipulate on the functional part through the pre-trained segmentation capability. We evaluate the effect of different pretraining methods for PointNet on the generalization and verify its effectiveness in Sec. IV.

Second, we design a task representation module to extract LTL representations to guide the agent to improve sampling efficiency. Compared with the traditional ways of combining automata and RL, encoding LTL instructions to extract

task semantics through Transformer not only builds the product-MDP on-the-fly to prevent the complexity of the algorithm from increasing exponentially with the LTL task, but also further improves the agent’s learning efficiency and manipulation ability through flexible characterization.

In addition, we design a contact planner based on affordance learning, which eliminates the need for offline data collection, as required in [11], making it more compatible with reinforcement learning. Our approach predicts the affordance map using the object’s point cloud and plans the optimal contact points o_{MP} for manipulation through an end-to-end learning framework. Consequently, the agent is guided effectively in manipulating articulated objects by leveraging both the optimal contact point o_{MP} and the designed reward r_{MPR} , which incentivizes the agent to reach the o_{MP} .

Unlike the computationally intensive setup of Meta-RL and the unimodal policy representation of traditional RL algorithms, we adopt a diffusion-based policy as the RL backbone for interacting with the environment. This approach addresses the challenges of using DM for policy representation, such as training from an offline RL perspective or learning from demonstrations. Furthermore, the convergence of the diffusion-based policy provides a theoretical foundation for ensuring both the reliability of action improvement and the effective generalization of DART, as demonstrated in Sec. IV.

Compared to other generalization methods discussed in Sec. I, the three modules in DART are tightly integrated, leading to a threefold improvement. Specifically, the LTL representation and MPO guide the diffusion-based policy to generate actions that maximize rewards. Effective interaction with articulated objects enhances the contact planner’s ability to predict more accurate MPO, while the resulting actions further refine the Transformer, leading to improved task semantics.

IV. EXPERIMENTS

In this section, we examine DART in comparison to previous works across the following aspects: 1) Performance: evaluating whether DART achieves higher success rates on both training and test sets in various scenarios. 2) Affordance: assessing the role of affordance guidance within DART’s framework. 3) Representation: analyzing how LTL representation directs the agent in executing dexterous manipulation of articulated objects. 4) Robustness: if DART maintains a high success rate when the camera view deviates from the training configuration.

A. Experimental Setup

1) *Environments*: DexArt [9], a benchmark for dexterous manipulation of various articulated objects with high degrees of freedom (DOF), is employed. Specific details about the reward function can be found on our website.

2) *Baselines*: To demonstrate the effectiveness of the DART framework, we empirically compare it to the following baselines. The first baseline is DexArt from [9]. Since a possible way to improve DexArt’s generalization performance is to leverage affordance learning to help the robot understand the task and thus improve the generalization, we extend DexArt with affordance learning (DexArt_{aff}) in the experiment. The

third baseline is QVPO [33], which leverages the exploration capabilities and multimodality of diffusion policies to improve its sample efficiency during online interaction. The fourth one is QSM [34], which differentiates through the denoising model and achieves policy convergence via Q-score matching, inherently enabling multimodal exploration in continuous domains. In addition, we add the baseline No Pretrain from [9] to represent the importance of pre-training for PointNet.

The above baselines and DART are performed on the Ubuntu 20.04 desktop with Intel Core i9 CPU, NVIDIA 3090 GPU and 32GB RAM. More details about baselines and hyperparameters are shown in website.

B. Experimental Results

1) *Performance Evaluation*: we evaluate the success rates of various methods across three scenarios, presenting the results in Table I. From the table, several observations can be made. (1) The method of extracting visual features via pre-training PointNet achieves a higher success rate compared to non-pre-training. (2) With the help of affordance learning, DART and DexArt_{aff} can get higher success rates than other methods without it. (3) QSM achieves better generalization performance relative to QVPO, which may be attributed to the better matching of the object’s pose distribution by Q-score and fewer denoising steps affecting the ability of QVPO in the generalization task. (4) By understanding the task semantics from LTL representations and MPO guidance, DART shows better results relative to other diffusion-based policy representation methods (QVPO and QSM). (5) DART demonstrates superior performance in most tasks, showing a higher success rate compared to DexArt_{aff} and DexArt. This advantage arises because DART employs a diffusion-based policy for decision-making, and its multi-modal policy representation enhances the agent’s performance in complex articulated object training environments. Notably, DART shows an approximate 45% improvement in solving the Unseen set of the Laptop task compared to DexArt..

2) *Success Rate and Average Success Steps*: We also present the success rate (red) and average success steps (blue) of DART for various tasks during training in Fig. 4. It can be observed that DART tends to converge at approximately half the total number of training steps, achieving a higher success rate and requiring fewer steps to complete the tasks.

3) *Pre-training of PointNet*: The segmentation pre-training method described in Sec. III-B is used to extract point cloud features and provide the agent with fundamental reasoning about the functional parts of articulated objects. To evaluate the effect of different pre-training methods on PointNet, we also consider Reconstruction [35] and SimSiam [36], presenting their corresponding success rates in Table II. From the table, we observe that 1) the pre-trained PointNet method using segmentation exhibits the highest success rate across all three scenarios. 2) Compared to other tasks, the segmentation method achieves a higher success rate in Laptop task.

C. Ablation Study

1) *The impact of different components*: We design the following ablation study to evaluate the influence of different

TABLE I
SUCCESS RATE OF DIFFERENT METHODS ON SIX SCENARIOS FOR BOTH SEEN AND UNSEEN OBJECTS.

Task	Toilet		Faucet		Laptop	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
No Pretrian	0.710 $\pm$ 0.050	0.460 $\pm$ 0.020	0.300 $\pm$ 0.220	0.280 $\pm$ 0.210	0.810 $\pm$ 0.010	0.490 $\pm$ 0.090
DexArt	0.850 $\pm$ 0.010	0.550 $\pm$ 0.010	0.790 $\pm$ 0.020	0.580 $\pm$ 0.070	0.920 $\pm$ 0.020	0.600 $\pm$ 0.070
DexArt _{aff}	0.898 $\pm$ 0.051	0.597 $\pm$ 0.063	0.760 $\pm$ 0.060	0.624 $\pm$ 0.182	0.542 $\pm$ 0.048	0.533 $\pm$ 0.000
QVPO	0.306 $\pm$ 0.199	0.291 $\pm$ 0.102	0.118 $\pm$ 0.092	0.095 $\pm$ 0.071	0.502 $\pm$ 0.028	0.472 $\pm$ 0.051
QSM	0.337 $\pm$ 0.092	0.208 $\pm$ 0.046	0.405 $\pm$ 0.056	0.213 $\pm$ 0.091	0.396 $\pm$ 0.272	0.383 $\pm$ 0.258
DART	0.944 $\pm$ 0.009	0.626 $\pm$ 0.014	0.836 $\pm$ 0.034	0.587 $\pm$ 0.022	0.952 $\pm$ 0.014	0.872 $\pm$ 0.054
Task	Trashcan		Dispenser		Box	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
No Pretrian	0.416 $\pm$ 0.288	0.261 $\pm$ 0.184	0.730 $\pm$ 0.049	0.366 $\pm$ 0.109	0.299 $\pm$ 0.050	0.294 $\pm$ 0.088
DexArt	0.503 $\pm$ 0.122	0.411 $\pm$ 0.169	0.783 $\pm$ 0.057	0.283 $\pm$ 0.061	0.780 $\pm$ 0.027	0.455 $\pm$ 0.061
DexArt _{aff}	0.680 $\pm$ 0.058	0.583 $\pm$ 0.040	0.776 $\pm$ 0.084	0.333 $\pm$ 0.021	0.404 $\pm$ 0.035	0.333 $\pm$ 0.108
QVPO	0.626 $\pm$ 0.092	0.511 $\pm$ 0.144	0.779 $\pm$ 0.021	0.362 $\pm$ 0.040	0.213 $\pm$ 0.118	0.316 $\pm$ 0.089
QSM	0.643 $\pm$ 0.040	0.516 $\pm$ 0.027	0.820 $\pm$ 0.000	0.375 $\pm$ 0.066	0.023 $\pm$ 0.006	0.061 $\pm$ 0.054
DART	0.733 $\pm$ 0.040	0.633 $\pm$ 0.026	0.880 $\pm$ 0.042	0.420 $\pm$ 0.021	0.861 $\pm$ 0.006	0.710 $\pm$ 0.020

TABLE II
SUCCESS RATE OF DIFFERENT VISUAL FEATURE EXTRACTION VIEWS ON FOUR SCENARIOS FOR BOTH SEEN AND UNSEEN OBJECTS.

Task	Toilet		Faucet		Laptop	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
No Pretrian	0.710 $\pm$ 0.050	0.460 $\pm$ 0.020	0.300 $\pm$ 0.220	0.280 $\pm$ 0.210	0.810 $\pm$ 0.010	0.490 $\pm$ 0.090
Reconstruction [35]	0.760 $\pm$ 0.030	0.520 $\pm$ 0.030	0.350 $\pm$ 0.020	0.210 $\pm$ 0.030	0.850 $\pm$ 0.040	0.540 $\pm$ 0.080
SimSiam [36]	0.820 $\pm$ 0.020	0.500 $\pm$ 0.060	0.600 $\pm$ 0.150	0.450 $\pm$ 0.120	0.840 $\pm$ 0.040	0.490 $\pm$ 0.130
Segmentation	0.944 $\pm$ 0.009	0.626 $\pm$ 0.014	0.836 $\pm$ 0.034	0.587 $\pm$ 0.022	0.952 $\pm$ 0.014	0.872 $\pm$ 0.054
Task	Trashcan		Dispenser		Box	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
No Pretrian	0.416 $\pm$ 0.288	0.261 $\pm$ 0.184	0.730 $\pm$ 0.049	0.445 $\pm$ 0.006	0.299 $\pm$ 0.050	0.294 $\pm$ 0.088
Reconstruction [35]	0.323 $\pm$ 0.232	0.182 $\pm$ 0.124	0.736 $\pm$ 0.028	0.408 $\pm$ 0.021	0.642 $\pm$ 0.084	0.655 $\pm$ 0.126
SimSiam [36]	0.470 $\pm$ 0.243	0.372 $\pm$ 0.177	0.739 $\pm$ 0.008	0.350 $\pm$ 0.040	0.537 $\pm$ 0.070	0.638 $\pm$ 0.061
Segmentation	0.733 $\pm$ 0.040	0.633 $\pm$ 0.026	0.880 $\pm$ 0.042	0.420 $\pm$ 0.021	0.861 $\pm$ 0.006	0.710 $\pm$ 0.020

TABLE III
SUCCESS RATES CORRESPONDING TO ABLATING DIFFERENT DART COMPONENTS ON SIX SCENARIOS.

Task	Toilet		Faucet		Laptop	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
DART _{w/o} ^{aff+LTL}	0.810 $\pm$ 0.084	0.461 $\pm$ 0.067	0.628 $\pm$ 0.118	0.330 $\pm$ 0.114	0.840 $\pm$ 0.068	0.731 $\pm$ 0.115
DART _{w/o} ^{aff}	0.863 $\pm$ 0.033	0.537 $\pm$ 0.018	0.716 $\pm$ 0.137	0.456 $\pm$ 0.099	0.922 $\pm$ 0.038	0.826 $\pm$ 0.037
DART _{w/o} ^{LTL}	0.715 $\pm$ 0.035	0.433 $\pm$ 0.022	0.675 $\pm$ 0.092	0.388 $\pm$ 0.126	0.948 $\pm$ 0.022	0.863 $\pm$ 0.029
DART	0.944 $\pm$ 0.009	0.626 $\pm$ 0.014	0.836 $\pm$ 0.034	0.587 $\pm$ 0.022	0.952 $\pm$ 0.014	0.872 $\pm$ 0.054
Task	Trashcan		Dispenser		Box	
Split	Seen	Unseen	Seen	Unseen	Seen	Unseen
DART _{w/o} ^{aff+LTL}	0.683 $\pm$ 0.073	0.588 $\pm$ 0.028	0.846 $\pm$ 0.067	0.308 $\pm$ 0.021	0.466 $\pm$ 0.013	0.616 $\pm$ 0.013
DART _{w/o} ^{aff}	0.716 $\pm$ 0.032	0.621 $\pm$ 0.102	0.840 $\pm$ 0.035	0.404 $\pm$ 0.005	0.571 $\pm$ 0.046	0.599 $\pm$ 0.062
DART _{w/o} ^{LTL}	0.726 $\pm$ 0.018	0.625 $\pm$ 0.042	0.656 $\pm$ 0.033	0.125 $\pm$ 0.020	0.694 $\pm$ 0.088	0.511 $\pm$ 0.083
DART	0.733 $\pm$ 0.040	0.633 $\pm$ 0.026	0.880 $\pm$ 0.042	0.420 $\pm$ 0.021	0.861 $\pm$ 0.006	0.710 $\pm$ 0.020

components in DART. (1) DART_{w/o}^{aff+LTL} is DART without considering affordance learning and LTL representations. (2) DART_{w/o}^{aff} is DART without considering affordance learning to show the importance of MPO o_{MP} and MPR r_{MPR} . (3) DART_{w/o}^{LTL} is DART without considering LTL representations to show the importance of task semantics understanding.

As shown in Table III, several observations can be made. (a) Without the help of the affordance learning and LTL representations, DART_{w/o}^{aff+LTL} get the worse performance on all scenarios. (b) By representing the LTL specifications, DART achieves better generalization results compared to DART_{w/o}^{LTL}. In the task representation module, the design of

a Transformer Encoder to encode LTL instructions not only facilitates the backpropagation of neural networks but also leverages the extracted task semantics. (c) By predicting MPO from the contact planner to guide the agent, DART shows better generalization performance compared to DART_{w/o}^{aff}.

2) *Key reward components visualization*: As shown in Fig. 5, we present the training process of the key reward components of DART in the Toilet task. It can be observed that (1) the increasing trend of MPR is similar to the curve of success rate, which means that the increasing MPR contributes to the success rate of the manipulation; (2) the articulation reward stabilizes around three million steps, which represents that the

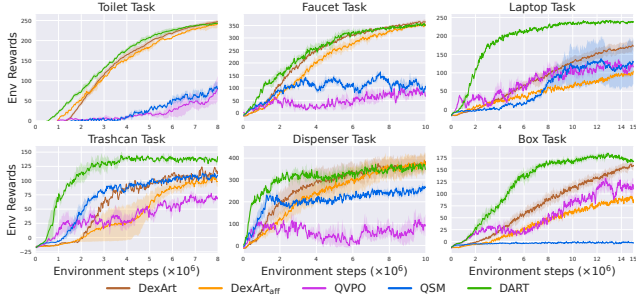


Fig. 3. Different Methods learning performing in seen objects of scenarios.

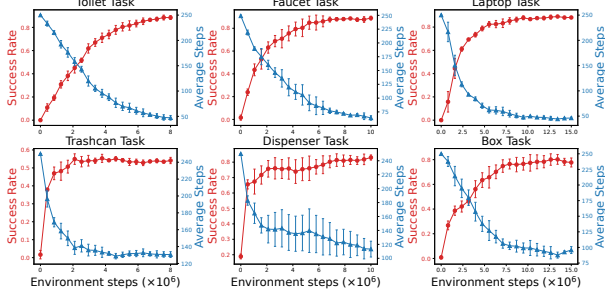


Fig. 4. Success rate (Red) and average success steps (blue) of DART for different tasks during training.

agent can succeed in most of the Toilet objects since that; and (3) the reach reward increases with the training process, which shows that the agent uses fewer steps to quickly approach the articulated object within a certain number of steps.

3) *The impact of task representation:* To demonstrate the effect of task representation on DART, we compare the performance of the following five task representation methods in DART in the Faucet task. The first method is DFA [37], which converts the task specification into the corresponding automata at the task level to fire critical task states to guide the agent. The second and third methods are LSTM and GRU, respectively, which effectively capture task information over time as text-based methods. The fourth is R-GCN [16], which utilizes explicit relational and structural information in the graph helping the agent to show better generalization ability.

TABLE IV
SUCCESS RATE OF DIFFERENT TASK REPRESENTATION METHODS.

Task Split	Faucet		Dispenser
	Seen	Unseen	Unseen (Transfer)
DFA	0.496 $\pm$ 0.038	0.347 $\pm$ 0.088	0.266 $\pm$ 0.056
LSTM	0.621 $\pm$ 0.100	0.409 $\pm$ 0.099	0.337 $\pm$ 0.056
GRU	0.550 $\pm$ 0.193	0.442 $\pm$ 0.141	0.266 $\pm$ 0.092
R-GCN	0.433 $\pm$ 0.118	0.290 $\pm$ 0.059	0.257 $\pm$ 0.065
Transformer (ours)	0.933 $\pm$ 0.010	0.639 $\pm$ 0.026	0.400 $\pm$ 0.044

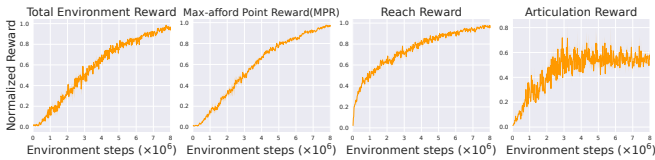


Fig. 5. Normalized reward components for DART in the Toilet task .

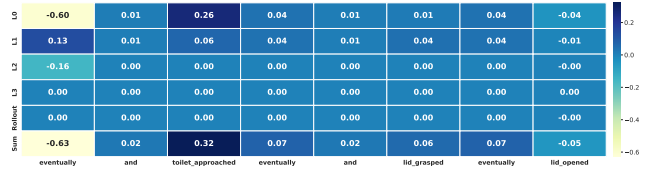


Fig. 6. The heatmap of the task φ_{toilet} by normalizing impact scores from different Transformer layers.

TABLE V
TOTAL TRAINING TIME (HOURS).

Task	Toilet	Faucet	Laptop	Trashcan	Dispenser	Box
DexArt	10.763	15.338	24.509	12.277	14.854	23.148
DexArt _{aff}	11.055	15.605	25.252	12.484	15.605	23.946
DART	11.883	15.432	23.020	11.574	14.172	22.281

And the last one is our method.

As shown in Table IV, it can be observed that 1) the method of encoding LTL instructions through Transformer achieves the highest success rate for the training set, the unseen set, and for the different category; 2) when using R-GCN as the task representation, the agent achieves the worst generalization performance, and it may be due to the over-smoothing by designing multiple message-passing layers; 3) the agent shows the second best task performance in all scenarios by exploiting the text-based method as the task representation, which means that text-based task representation can effectively capture long-term dependencies and understand the context within tasks.

4) *The visualization of task representation:* To further illustrate the agent's understanding of the LTL task, Fig. 6 illustrates the heatmap of the task φ_{toilet} by exploiting AttCAT [38]. Upon the convergence of Transformer, higher impact scores from all layers focus on the token `toilet_approached` (+0.32) as shown in Fig. 6, which suggests that the agent is more likely to move directly to the position corresponding to the `toilet_approached` proposition.

5) *The comparison of time consumption:* To further clarify the training efficiency of DART at the time level, we evaluate the average time consumed by DexArt, DexArt_{aff} and DART on the above six tasks. As shown in Fig. 3 and Table. V, we observe that 1) when the tasks are simple (e.g., the Toilet and Faucet tasks), the three methods exhibit similar reward curves, and DexArt consumes the least average time due to its simplicity. 2) When the tasks are more challenging, DART not only demonstrates efficient learning performance, but also uses the shortest training time.

D. Transfer Reasoning of Unseen Categories

In [9], a dexterous manipulation task benchmark is proposed to evaluate the ability of generalizing across articulated objects within the same category. We believe that an algorithm capable of effectively reasoning about the functional parts of articulated objects should be able to generalize well to objects within the same category and also exhibit the ability to generalize to objects across different categories. Therefore, we extend the categories of articulated objects beyond those in [9], as detailed on our website, to evaluate the generalization performance across different categories.

TABLE VI
SUCCESS RATE EVALUATION OF DIFFERENT METHODS WHEN
TRANSFERING TO DIFFERENT CATEGORY-LEVEL OBJECTS.

Task (Full)	Trashcan (Toilet)	Dispenser (Faucet)	Box (Laptop)
DexArt	0.364 $\pm$ 0.122	0.246 $\pm$ 0.014	0.178 $\pm$ 0.003
DexArt _{aff}	0.447 $\pm$ 0.060	0.438 $\pm$ 0.031	0.243 $\pm$ 0.243
QVPO	0.210 $\pm$ 0.129	0.353 $\pm$ 0.103	0.256 $\pm$ 0.031
QSM	0.224 $\pm$ 0.088	0.434 $\pm$ 0.081	0.222 $\pm$ 0.100
DART	0.562 $\pm$ 0.051	0.518 $\pm$ 0.025	0.435 $\pm$ 0.072
Task (Heterogeneous)	Trashcan (Toilet)	Dispenser (Faucet)	Box (Laptop)
DexArt	0.393 $\pm$ 0.031	0.241 $\pm$ 0.025	0.144 $\pm$ 0.008
DexArt _{aff}	0.388 $\pm$ 0.064	0.233 $\pm$ 0.062	0.199 $\pm$ 0.084
QVPO	0.127 $\pm$ 0.034	0.138 $\pm$ 0.056	0.044 $\pm$ 0.031
QSM	0.038 $\pm$ 0.020	0.349 $\pm$ 0.056	0.077 $\pm$ 0.039
DART	0.472 $\pm$ 0.103	0.400 $\pm$ 0.044	0.361 $\pm$ 0.088
Task (Complex)	Trashcan (Toilet)	Dispenser (Faucet)	Box (Laptop)
DexArt	0.354 $\pm$ 0.032	0.234 $\pm$ 0.005	0.145 $\pm$ 0.027
DexArt _{aff}	0.320 $\pm$ 0.054	0.321 $\pm$ 0.039	0.153 $\pm$ 0.022
QVPO	0.085 $\pm$ 0.052	0.271 $\pm$ 0.037	0.194 $\pm$ 0.036
QSM	0.078 $\pm$ 0.051	0.328 $\pm$ 0.038	0.148 $\pm$ 0.050
DART	0.406 $\pm$ 0.026	0.395 $\pm$ 0.029	0.281 $\pm$ 0.036

Taking the Trashcan task as an example, we train the agent on the set of seen objects in the Toilet task, and further test its success rate for unseen object types in Trashcan task, denoted as Trashcan (Toilet). The same applies to Faucet (Dispenser) and Box (Laptop). We evaluate the success rate of different methods when transferring to different category-level objects and show the results in Table VI.

From Table VI, it can be observed that 1) compared with Table I, DART still shows good generalization even different category-level objects; 2) Although DART shows lower success rates in Heterogeneous and Complex sets, compared with other algorithms, it still has a clear lead advantage. 3) Although QSM and QVPO are also diffusion-based policy representation methods, they have shown better results only on the Dispenser and Trashcan tasks, which may be due to the capacity limitation of Q-Score matching and the small number of diffusion steps; 4) By predicting MPO through the contact planner, affordance learning-based algorithms (DexArt_{aff} and DART) generally achieve a good success rate; 5) DART shows the highest success rate in all scenarios, and this phenomenon confirms that the generalization of diffusion-based policy can be further improved under the guidance of task semantics and MPO. The snapshots for one of different category-level generalizations are shown in Fig. 7.

E. Robustness to Viewpoint Change

We further test the robustness of DART when changing the camera viewpoints in Laptop task following the approach of [9]. The success rate for the unseen set corresponding to a total of 35 camera views is shown in Fig. 8.

As shown in Fig. 8, DexArt, which already exhibits a low success rate on the unseen set, shows a significant decline in success rate as the azimuthal angle changes, reaching only about 10%. In contrast, DART demonstrates much more robust performance, maintaining a success rate of over 70% across all 35 views, and achieving over 85% success for azimuths ranging from -20° to 60° . Additionally, while the training view is the optimal manipulation view for DexArt, DART manages to achieve higher success rates than the training view even as the azimuthal and polar angles vary.

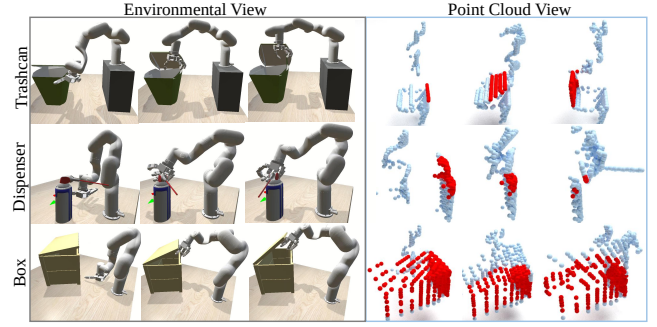


Fig. 7. Snapshots from the environment views and the point cloud views. The completed point cloud views are shown in our website.

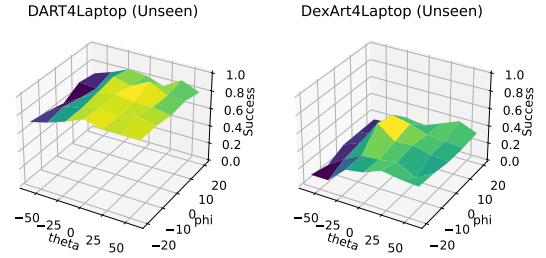


Fig. 8. We evaluate Success Rate on Laptop task under Different Viewpoints. The x -axis is the polar angle ϕ (relative to the training viewpoint) and the y -axis is the azimuthal angle θ on the semi-sphere centered at the object. The z -axis represents the success rate.

F. Real-World Experimental Results

To further validate the generalization of DART, physical experiments were conducted, as illustrated in Fig. 9. We utilize Realsense D455 to capture point clouds, and exploit the UR10e arm and Shadow Hand to execute the action output by different methods. The corresponding results are all performed on the Ubuntu 20.04 laptop with Intel Core i7-12800HX CPU, NVIDIA 3080TI GPU and 32GB RAM. We train the performance of the UR10e arm and Shadow Hand in the Laptop task in simulation, and deploy the corresponding weights directly in the real world to evaluate the performance that the weights have. It is worth noting that we do not further adapt or optimize the corresponding weights during deployment, and the observed noise in the environment will challenge the performance of the algorithm, thus further demonstrating the robustness and generalizability of DART. We first assess the performance of DART with two laptops and a box. The snapshots and corresponding point cloud views are shown in Fig. 10. And we then evaluate the success of different methods by randomly initializing the position and orientation on the Laptop task as shown in Table VII. A full video of the experiment is available on our website.

1) *Manipulation effectiveness*: It can be observed that (1) DexArt and DexArt_{aff} have a low success rate because, under the same seed, although they have a certain success rate in the simulation, they have to be terminated artificially early due to the dangerous nature of the corresponding actions in the real world. (2) QVPO has poor performance in real-world scenarios because it is difficult to succeed in simulation evaluation. (3) QSM has a second better performance in the

TABLE VII
SUCCESS RATE EVALUATION AND SAFETY VIOLATION RATE FOR REAL
LAPTOP EXPERIMENTS. EACH TASK IS EVALUATED WITH 10 TRIALS.

Laptop Task	DexArt	DexArt _{aff}	QVPO	QSM	DART
Laptop (Position)	0	20	0	40	80
Laptop (Orientation)	0	0	0	60	90
Average (Mean)	0	10	0	50	85
Safety Violation Rate	DexArt	DexArt _{aff}	QVPO	QSM	DART
Laptop (Position)	100	80	0	0	0
Laptop (Orientation)	100	100	0	0	0
Average (Mean)	100	90	0	0	0

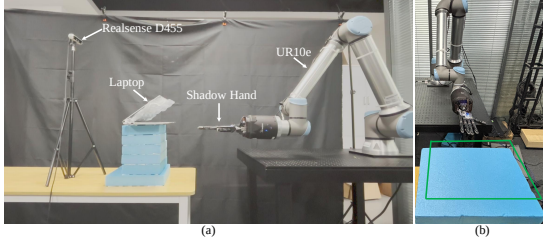


Fig. 9. (a) The real-world experimental platform constructed for Task Laptop. (b) The green frame means randomized object poses.

real-world scenarios by exploiting through Q-score matching and differentiating through the denoising model. (4) DART not only has the highest success rate in both scenarios, but also the actions are more anthropomorphic than other methods.

2) *Safety violation*: In real-world experiments, we observe that some methods may have unpredictable behaviors that must be terminated to ensure safety. We define this situation as safety violation and show the corresponding results in Table VII and Fig. 11. It can be observed that (1) the operations of DexArt and DexArt_{aff} are highly likely to cause damage to the robot due to the radical exploration of entropy; (2) the methods based on policy characterization (QVPO, QSM, and DART) although not necessarily having high success rates, exhibit actions that are consistently safe, possibly due to the diffusion-based policy performing K iterations to gradually denoise a random action into the noise-free action, which are not only robust to changes in the environment but also synthesize multimodal features of action distributions to avoid selecting rare or extreme, potentially dangerous actions.

V. CONCLUSIONS

In this work, we introduce DART, which leverages LTL representations to enhance task semantics understanding and

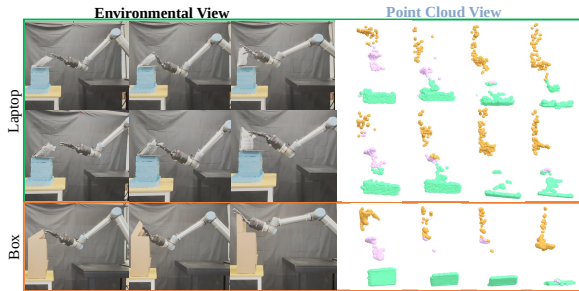


Fig. 10. Snapshots and the corresponding point cloud views for Task Laptop.



Fig. 11. Example of safety violation when evaluating DexArt and DexArt_{aff} for the real-world experiment.

improve the learning efficiency, affordance learning to guide efficient manipulation of articulated objects based on the predicted max-affordance point observation, and a diffusion-based policy to improve generalization across diverse object types. We enhance the action optimization method to address challenges inherent in traditional diffusion policies that rely solely on prior information. Extended experiments demonstrate the superior performance of DART over existing methods.

Despite its great potential, several challenges remain. First, when designing the diffusion-based policy in DART, we did not extend it to the classifier-guided case. This is because according to the experiments in [39] and [40], incorporating classifier-guided samples in the RL setting may degrade algorithm performance, even though it is a useful technique in computer vision. Another potential issue is that the Task Representation Module is updated indirectly through the gradient of the diffusion-based policy. Finding more ways to directly update the LTL encoder by utilizing rewards may further improve the performance of the algorithm. Hence, future research will consider leveraging the policy with the classifier-guided diffusion to tackle more intricate dexterous manipulation tasks, such as tool understanding and utilization.

REFERENCES

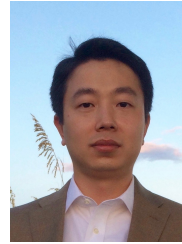
- [1] S. Wang, W. Hu, L. Sun, X. Wang, and Z. Li, "Learning adaptive grasping from human demonstrations," *IEEE ASME Trans Mechatron*, vol. 27, no. 5, pp. 3865–3873, 2022.
- [2] Z. Xu, Z. He, and S. Song, "Universal manipulation policy network for articulated objects," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 2447–2454, 2022.
- [3] Z. Lu, N. Wang, and C. Yang, "A constrained dmps framework for robot skills learning and generalization from human demonstrations," *IEEE ASME Trans Mechatron*, vol. 26, no. 6, pp. 3265–3275, 2021.
- [4] H. Wang, H. Zhang, L. Li, Z. Kan, and Y. Song, "Task-driven reinforcement learning with action primitives for long-horizon manipulation skills," *IEEE Trans. Cybern.*, 2023.
- [5] K. Rakelly, A. Zhou, C. Finn, S. Levine, and D. Quillen, "Efficient off-policy meta-reinforcement learning via probabilistic context variables," in *Int. Conf. Mach. Learn.* PMLR, 2019, pp. 5331–5340.
- [6] D. Yarats, R. Fergus, A. Lazaric, and L. Pinto, "Mastering visual continuous control: Improved data-augmented reinforcement learning," in *Int. Conf. Learn. Represent., ICLR*, 2022.
- [7] S. Nair, A. Rajeswaran, V. Kumar, C. Finn, and A. Gupta, "R3m: A universal visual representation for robot manipulation," in *Proc. Mach. Learn. Res.* PMLR, 2022, pp. 892–909.
- [8] K. Gmelin, S. Bahl, R. Mendonca, and D. Pathak, "Efficient RL via disentangled environment and agent representations," in *Int. Conf. Mach. Learn.* PMLR, 2023, pp. 11 525–11 545.
- [9] C. Bao, H. Xu, Y. Qin, and X. Wang, "Dexart: Benchmarking generalizable dexterous manipulation with articulated objects," in *Proc. - IEEE Conf. Comput. Vis. Pattern Recognit, CVPR*, 2023, pp. 21 190–21 200.
- [10] J. J. Gibson, "The theory of affordances," *Hilldale, USA*, vol. 1, no. 2, pp. 67–82, 1977.
- [11] K. Mo, L. J. Guibas, M. Mukadam, A. Gupta, and S. Tulsiani, "Where2act: From pixels to actions for articulated 3d objects," in *Proc. IEEE Int. Conf. Comput. Vision.*, 2021, pp. 6813–6823.

- [12] R. Wu, C. Ning, and H. Dong, "Learning foresightful dense visual affordance for deformable object manipulation," in *Proc. IEEE Int. Conf. Comput. Vision.*, 2023, pp. 10947–10956.
- [13] Y. Geng, B. An, H. Geng, Y. Chen, Y. Yang, and H. Dong, "Rlafford: End-to-end affordance learning for robotic manipulation," in *Proc. IEEE Int. Conf. Rob. Autom.*, 2023, pp. 5880–5886.
- [14] C. Baier and J.-P. Katoen, *Principles of model checking*. MIT press, 2008.
- [15] X. Li, Z. Serlin, G. Yang, and C. Belta, "A formal methods approach to interpretable reinforcement learning for robotic planning," *Sci. Robot.*, vol. 4, no. 37, 2019.
- [16] P. Vaezipoor, A. C. Li, R. A. T. Icarte, and S. A. McIlraith, "Ltl2action: Generalizing LTL instructions for multi-task RL," in *Proc. Int. Conf. Machin. Learn.*. PMLR, 2021, pp. 10497–10508.
- [17] H. Zhang, H. Wang, and Z. Kan, "Exploiting transformer in sparse reward reinforcement learning for interpretable temporal logic motion planning," *IEEE Robot. Autom. Lett.*, 2023.
- [18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," *Adv. neural inf. process. syst.*, vol. 30, 2017.
- [19] J. Wang, E. Shi, H. Hu, C. Ma, Y. Liu, X. Wang, Y. Yao, X. Liu, B. Ge, and S. Zhang, "Large language models for robotics: Opportunities, challenges, and perspectives," *Journal of Automation and Intelligence*, 2024.
- [20] D. Jung, C. Gu, J. Park, and J. Cheong, "Touch gesture recognition-based physical human-robot interaction for collaborative tasks," *IEEE Trans. Cogn. Dev. Syst.*, 2024.
- [21] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Int. Conf. Mach. Learn.*. PMLR, 2015, pp. 2256–2265.
- [22] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 6840–6851, 2020.
- [23] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *Int. Conf. Learn. Represent., ICLR*, 2021.
- [24] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Robot. Sci. Syst., RSS*, 2023.
- [25] L. Yang, Z. Huang, F. Lei, Y. Zhong, Y. Yang, C. Fang, S. Wen, B. Zhou, and Z. Lin, "Policy representation via diffusion probability model for reinforcement learning," *arXiv preprint arXiv:2305.13122*, 2023.
- [26] D.-M. Shi, T. Huang, X.-M. Zhang, and H. Ding, "Superresolution restoration of cutting forces in thin-walled milling with a diffusion model," *IEEE ASME Trans Mechatron*, 2024.
- [27] O. Kupferman and M. Y. Vardi, "Model checking of safety properties," *Form. Methods Syst. Des.*, vol. 19, no. 3, pp. 291–314, 2001.
- [28] M. Tuli, A. Li, P. Vaezipoor, T. Klassen, S. Sanner, and S. McIlraith, "Learning to follow instructions in text-based games," *Adv. Neural Inf. Process. Syst.*, vol. 35, pp. 19441–19455, 2022.
- [29] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," in *Proc. - IEEE Conf. Comput. Vis. Pattern Recognit, CVPR*, 2017, pp. 652–660.
- [30] J. Borja-Diaz, O. Mees, G. Kalweit, L. Hermann, J. Boedecker, and W. Burgard, "Affordance learning from play for sample-efficient policy learning," in *Proc. IEEE Int. Conf. Rob. Autom.*. IEEE, 2022, pp. 6372–6378.
- [31] J. Jian, X. Liu, M. Li, R. Hu, and J. Liu, "Affordpose: A large-scale dataset of hand-object interactions with affordance-driven hand pose," in *Proc. IEEE Int. Conf. Comput. Vision.*, 2023, pp. 14713–14724.
- [32] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "Pointnet++: Deep hierarchical feature learning on point sets in a metric space," *Adv. Neural Inf. Process. Syst.*, vol. 30, pp. 5099–5108, 2017.
- [33] S. Ding, K. Hu, Z. Zhang, K. Ren, W. Zhang, J. W. Jingyi Yu, and Y. Shi, "Diffusion-based reinforcement learning via q-weighted variational policy optimization," *Adv. neural inf. process. systs.*, vol. 37, pp. 53945–53968, 2024.
- [34] M. Psenka, A. Escontrela, P. Abbeel, and Y. Ma, "Learning a diffusion model policy from rewards via q-score matching," in *Proc. Int. Conf. Machin. Learn.*. PMLR, 2024, pp. 41163–41182.
- [35] J. Xie, Y. Xu, Z. Zheng, S.-C. Zhu, and Y. N. Wu, "Generative pointnet: Deep energy-based learning on unordered point sets for 3d generation, reconstruction and classification," in *Proc. - IEEE Conf. Comput. Vis. Pattern Recognit, CVPR*, 2021, pp. 14976–14985.
- [36] X. Chen and K. He, "Exploring simple siamese representation learning," in *Proc. - IEEE Conf. Comput. Vis. Pattern Recognit, CVPR*, 2021, pp. 15750–15758.
- [37] B. Lacerda, D. Parker, and N. Hawes, "Optimal and dynamic planning for markov decision processes with co-safe LTL specifications," in *IEEE/RSJ Int. Conf. Intell. Robot. Syst.*. IEEE, 2014, pp. 1511–1516.
- [38] Y. Qiang, D. Pan, C. Li, X. Li, R. Jang, and D. Zhu, "Attcat: Explaining transformers via attentive class activation tokens," *Adv. Neural Inf. Process. Syst.*, vol. 35, pp. 5052–5064, 2022.
- [39] T. Pearce, T. Rashid, A. Kanervisto, D. Bignell, M. Sun, R. Georgescu, S. V. Macua, S. Z. Tan, I. Momennejad, K. Hofmann *et al.*, "Imitating human behaviour with diffusion models," in *Int. Conf. Learn. Represent., ICLR*, 2023.
- [40] M. Reuss, M. Li, X. Jia, and R. Lioutikov, "Goal-conditioned imitation learning using score-based diffusion policies," in *Robot. Sci. Syst., RSS*, 2023.



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